

Who Pays the Bill?

Substrate Skepticism, Stable Counterfactual Identity,
Hidden Synergy, and Information-Spectrum Geometry

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Version 3

Abstract

This note asks a deliberately contract-dependent question. Suppose an observable system is specified only by controlled stochastic input–output laws. Without identifying the underlying ontology, what information resources must an exact realization carry once one specifies which mutually exclusive possibilities are required to belong to one common latent construction?

We distinguish chosen-path sampling, one-step counterfactual coupling, a *synchronous* counterfactual contract in which the latent coordinate $F_t(a)$ has a stable identity across histories, and a more general history-indexed causal response tree $F_t(h_{t-1}, a)$. This distinction corrects an ambiguity in the previous version: exactness for every adaptive policy under the synchronous table does not imply that every causal response tree must admit the same history-blind representation.

For the synchronous contract we define the minimum block entropy C_n^{syn} . Its transversal constraints imply exactness for every adaptive nonanticipative policy. We prove an algebraic rank formula for the corresponding marginal polytope, give a general perturbative mechanism for hidden synergy, and exhibit an improved 23-atom two-step witness showing strict block subadditivity. For the equal-entropy three-outcome crash test,

$$C_2^{\text{syn}} \leq 3.8782955762 < 2C_1,$$

so mutually exclusive futures can share latent information and reduce the bill.

The main converse is generalized from two actions to an arbitrary finite action family. Let μ_a be the law of the one-symbol surprisal $-\log_2 P_a(Y)$. Every synchronous block obeys

$$C_n^{\text{syn}}(P_a : a \in \mathcal{A}) \geq n \int_0^1 \max_{a \in \mathcal{A}} F_{\mu_a}^{-1}(u) du.$$

For two actions this reduces to

$$C_n^{\text{syn}}(P, Q) \geq \frac{n}{2} (H(P) + H(Q) + W_1(\mu_P, \mu_Q)).$$

For the crash-test pair this yields the certified asymptotic interval

$$1.7284654687 \leq C_\infty^{\text{syn}} \leq 1.9391477881 < C_1 = 2.0539185856.$$

Thus hidden correlations reduce the synchronous counterfactual bill but cannot erase the self-information-spectrum mismatch.

A previously written kernel note turns out to contain the exact local overlap appearing in the converse: for its $k = 0$ compact-support kernel,

$$2F_0(1/p, 1/q) = \min(p, q),$$

so in surprisal coordinates

$$-\log_2(2F_0(2^u, 2^v)) = \max(u, v).$$

This is an exact representation of the scalar building block, not a claim that the earlier note contained the counterfactual theorem.

Finally we define a general history-indexed causal-tree entropy C_n^{causal} . We prove

$$C_1 + (n - 1)h_{\max} \leq C_n^{\text{causal}} \leq C_n^{\text{syn}},$$

but we do not know whether $C_\infty^{\text{causal}} > h_{\max}$. The positive asymptotic tax proved here is therefore a theorem about stable synchronous counterfactual identity, while the cost of unrestricted causal counterfactual coherence remains open.

No historical novelty is claimed. The construction lies near minimum-entropy coupling, information-spectrum converses, common information, exact channel synthesis, sample-privacy synergy, d -wise independence, marginal polytopes, finitary coding, and causal optimal transport. The standing assumption is that an equivalent or stronger result may already exist under different terminology unless a dedicated literature search establishes otherwise.

“The accountant discovered netting. The auditor discovered a lower bound. The lawyer asked which contract we had signed.”

1 Priority convention and philosophical origin

1.1 Priority convention

The purpose of this note is to sharpen a resource question, not to manufacture a novelty claim. We therefore keep the standing convention of the previous versions.

Priority convention. Assume that any clean statement derived here has probably been stated, proved, or embedded in a more general theory somewhere else. The word *derived* describes only the internal logic of this note. When a nearby precedent is known, it is cited; when no direct precedent has been located, we say only that we have not located one.

This caution is substantive. Minimum-entropy coupling is established information theory [2]; information-spectrum converses and efficient approximate multi-marginal couplings already exist [3, 4]. Exact common information and exact channel synthesis have multi-letter rate characterizations [5, 6]. Ex-ante contingent plans versus local randomization have a long history in game theory [11, 12]. Strong structured-independence constraints have their own entropy theory [13]; hierarchical marginal polytopes connect naturally to coding geometry [14]. Causal and bicausal optimal transport explicitly distinguish ordinary couplings from couplings compatible with a filtration [15]. Sample-privacy synergy, all-or-nothing transforms, catalytic entropy, and finitary coding provide further neighboring mechanisms [16, 17, 18, 19, 20, 21, 22].

The exact combination of synchronous transversal constraints, minimum full-table entropy, and the specific converses below was not located in this formulation in the present search. This sentence is not a priority assertion.

1.2 From a dream to a rule

A philosophical trigger was Pavel Arazim’s public answer for the Czech outreach project *Zeptej se vědce* to the question “What if life is a dream and one wakes up after death?” [1]. The connection is motivational only and does not imply endorsement of the mathematics below.

The useful question is not whether the world is a dream. A single realized sequence is only a history. A rule begins when one constrains what would happen under alternative admissible interventions. We therefore ask

If several mutually exclusive histories are required to obey one common rule, what must that common realization carry?

This is the form of *substrate skepticism* used here. We do not infer what lies underneath. We specify a realization contract, then price it.

2 The accounting-standard principle

A recurrent failure mode in discussions of randomness, simulation, and “fundamental reality” is to move between inequivalent implementation contracts without noticing. We distinguish four levels.

C1. Chosen-path sampling. At time t the action A_t is already known. Only the requested output $Y_t \sim P_{A_t}$ must be produced.

C2. One-step counterfactual table. Before the action is selected, one latent draw determines

$$F_t = (F_t(a))_{a \in \mathcal{A}}, \quad F_t(a) \sim P_a.$$

Only one coordinate is observed, but all coordinates are jointly defined.

C3-S. Synchronous stable-event table. For a finite horizon, one latent source fixes

$$F = (F_t(a))_{1 \leq t \leq n, a \in \mathcal{A}}.$$

The coordinate indexed by the same (t, a) is the same latent event across all histories. This is the contract analyzed most sharply below.

C3-C. History-indexed causal tree. A more general latent plan fixes

$$F_t(h_{t-1}, a),$$

where the counterfactual response may depend on the complete prior history h_{t-1} . This is a weaker stable-identity requirement and is not automatically equivalent to C3-S.

Thus

The bill is meaningful only after the accounting standard is fixed.

The correction from Version 2 is important: C3-S realizes every adaptive policy exactly, but it is not the most general causal counterfactual contract.

3 Controlled channels and the one-shot bill

Let $\mathcal{A}$ and $\mathcal{Y}$ be finite. A memoryless controlled channel is a family

$$W = \{P_a : a \in \mathcal{A}\}, \quad P_a \in \Delta(\mathcal{Y}),$$

satisfying, for every adaptive action process,

$$\Pr(Y_t = y \mid A_{1:t}, Y_{1:t-1}) = P_{A_t}(y). \tag{3.1}$$

Define

$$h_{\max} := \max_{a \in \mathcal{A}} H(P_a). \tag{3.2}$$

The policy that always chooses an entropy-maximizing action forces at least $h_{\max}$ observable output entropy per step.

Under C2, a latent one-step table $F = (F(a))_{a \in \mathcal{A}}$ has prescribed coordinate marginals. Its minimum entropy is the multi-marginal minimum-entropy coupling

$$C_1(W) := \min\{H(F) : \text{Law}(F(a)) = P_a \ \forall a \in \mathcal{A}\} = \text{MEC}(P_a : a \in \mathcal{A}). \quad (3.3)$$

The one-shot surcharge

$$r_1(W) := C_1(W) - h_{\max} \geq 0 \quad (3.4)$$

is real. It is not automatically a per-step tax.

4 Chosen-path sampling nearly reaches Shannon

Under C1, modern exact sampling already defeats the naive idea that adaptive choice itself forces a positive entropy-rate surcharge. Randomness-recycling samplers approach the Shannon rate while retaining only logarithmic persistent state in the entropy slack [8]. The real-probability extension obtains exact adaptive sampling with only logarithmic total entropy overhead in the horizon [9], while complementary lower bounds quantify the memory cost of approaching Shannon uniformly [10].

Thus chosen-path randomness can be recycled almost to the observable floor. Any additional bill must come from a stronger requirement than merely producing the path that was actually requested.

5 Warm-up: a strict reset obstruction

To isolate one extreme accounting standard, let a finite reservoir $S \sim \lambda$ and a finite innovation $U \sim \nu$ be independent. A *strong reset recycler* for action a is a deterministic map

$$\Phi_a : S \times U \rightarrow \mathcal{Y} \times S$$

such that

$$\Phi_a(S, U) = (Y, S') \sim P_a \otimes \lambda. \quad (5.1)$$

The output and returned reservoir are therefore independent and the reservoir has exactly reset to its original law.

Proposition 5.1 (Entropy balance). *Every strong reset recycler satisfies*

$$H(\nu) \geq \max_a H(P_a). \quad (5.2)$$

Proof. For every action, (Y, S') is a deterministic function of (S, U) , so

$$H(P_a) + H(\lambda) = H(Y, S') \leq H(S, U) = H(\lambda) + H(\nu).$$

□

Theorem 5.2 (Exact reset floor forces spectral equivalence). *Let P, Q be finite probability vectors with*

$$H(P) = H(Q) = h.$$

If one finite strong reset recycler works for both actions with $H(\nu) = h$, then P and Q have the same multiset of nonzero atom probabilities, up to permutation and zero padding.

Proof. At equality, the deterministic map from (S, U) to (Y, S') preserves entropy and is therefore invertible almost surely on its support. Hence

$$\lambda \otimes \nu \equiv_{\text{perm}} P \otimes \lambda \quad \text{and} \quad \lambda \otimes \nu \equiv_{\text{perm}} Q \otimes \lambda.$$

Thus $P \otimes \lambda$ and $Q \otimes \lambda$ have identical atom multisets. Equality of every power sum gives

$$\left(\sum_i p_i^r\right) \left(\sum_j \lambda_j^r\right) = \left(\sum_i q_i^r\right) \left(\sum_j \lambda_j^r\right),$$

and the second factor is positive. Newton's identities then recover the atom multiset of P and Q . $\square$

The reset condition is intentionally strong. Correlated catalysis already warns that independence of the surviving auxiliary system can itself be a costly resource [18, 19]. The next sections remove reset independence entirely.

6 A concrete equal-entropy crash test

Consider

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \quad Q = (q, q, 1 - 2q), \quad (6.1)$$

where $q \in (1/3, 1/2)$ is the unique solution of

$$-2q \log_2 q - (1 - 2q) \log_2 (1 - 2q) = \frac{3}{2}. \quad (6.2)$$

Numerically,

$$q = 0.41003242818198557608\dots, \quad H(P) = H(Q) = 1.5 \text{ bits}. \quad (6.3)$$

The spectra are not permutations.

A direct vertex enumeration of the 3×3 transportation polytope gives a one-step minimum-entropy coupling with nonzero probabilities

$$\frac{1}{2} - q, \quad q, \quad 2q - \frac{3}{4}, \quad 1 - 2q, \quad \frac{1}{4}, \quad (6.4)$$

and hence

$$C_1(P, Q) = 2.0539185856072597\dots \text{ bits}. \quad (6.5)$$

7 Synchronous policy-safe blocks

For a block of length n , let

$$F = (F_t(a))_{1 \leq t \leq n, a \in \mathcal{A}} \quad (7.1)$$

be one jointly distributed synchronous counterfactual table.

Definition 7.1 (Synchronous policy-safe law). A law Γ of F is *synchronous policy-safe* if for every deterministic action string $a^n = (a_1, \dots, a_n)$,

$$\text{Law}_\Gamma(F_1(a_1), \dots, F_n(a_n)) = \bigotimes_{t=1}^n P_{a_t}. \quad (7.2)$$

Proposition 7.2 (Transversal criterion). *A synchronous table law is policy-safe if and only if it realizes the exact memoryless channel for every deterministic nonanticipative policy. The same holds for randomized policies with independent private randomization.*

Proof. Necessity follows from policies that ignore history. Conversely, fix a deterministic policy and an output string y^n . Along that branch the policy determines a fixed action string

$$a_t = a_t(y_{1:t-1}).$$

The branch event is

$$\bigcap_{t=1}^n \{F_t(a_t(y_{1:t-1})) = y_t\},$$

whose probability is the required product by policy-safety. Conditioning on independent private policy randomness gives the randomized case. $\square$

Define

$$C_n^{\text{syn}}(W) := \min\{H(F) : F \text{ is synchronous policy-safe}\}. \quad (7.3)$$

The feasible set is a nonempty compact polytope, so the minimum exists.

Proposition 7.3 (Subadditivity and a prefix lower bound). *For all $m, n \geq 1$,*

$$C_{m+n}^{\text{syn}} \leq C_m^{\text{syn}} + C_n^{\text{syn}}, \quad (7.4)$$

and

$$C_{m+n}^{\text{syn}} \geq mh_{\max} + C_n^{\text{syn}}. \quad (7.5)$$

Consequently

$$C_\infty^{\text{syn}} := \lim_{n \rightarrow \infty} \frac{C_n^{\text{syn}}}{n} = \inf_n \frac{C_n^{\text{syn}}}{n}, \quad (7.6)$$

and the excess sequence

$$E_n^{\text{syn}} := C_n^{\text{syn}} - nh_{\max} \quad (7.7)$$

is nondecreasing.

Proof. Independent concatenation gives subadditivity. For the lower bound, choose an entropy-maximizing action during the first m times and retain the entire future n -step table. This projection of the full table has entropy

$$mh_{\max} + H(F_{m+1:m+n} \mid Y_{1:m}).$$

Conditioned on each positive-probability prefix, the future table is again synchronous policy-safe, so its entropy is at least C_n^{syn} . $\square$

For the crash test,

$$C_2^{\text{syn}} \geq h_{\max} + C_1 = 3.5539185856072597 \dots \text{ bits}. \quad (7.8)$$

7.1 The algebraic skeleton of the transversal polytope

Suppose action a has output alphabet size d_a . At one time the full counterfactual state lies in

$$\mathcal{R} := \prod_{a \in \mathcal{A}} \mathcal{Y}_a.$$

Let V_a be the space of real functions on $\mathcal{R}$ that depend only on coordinate a , and let

$$\mathcal{O} := \sum_{a \in \mathcal{A}} V_a. \quad (7.9)$$

Proposition 7.4 (Observable-space dimension and block rank). *The one-time observable space has dimension*

$$d_{\text{obs}} = 1 + \sum_{a \in \mathcal{A}} (d_a - 1), \quad (7.10)$$

and the row space generated by all n -time transversal marginal constraints is

$$\mathcal{O}^{\otimes n}.$$

Hence the independent constraint rank is

$$\boxed{\text{rank } A_n = d_{\text{obs}}^n}. \quad (7.11)$$

Proof. A basis for $\mathcal{O}$ consists of one constant function together with, for each action, $d_a - 1$ independent coordinate contrasts. Across n times, a fixed action string generates a tensor product of the corresponding coordinate spaces. Taking the span over all action strings gives

$$\left(\sum_a V_a \right)^{\otimes n} = \mathcal{O}^{\otimes n}.$$

□

For two ternary actions, $|\mathcal{R}| = 9$ and $d_{\text{obs}} = 5$, so

$$\text{rank } A_n = 5^n, \quad \dim \ker A_n = 9^n - 5^n. \quad (7.12)$$

The kernel contains precisely the directions invisible to every allowed transversal constraint. This is the linear home in which hidden interaction can live. Hierarchical marginal polytopes and their relation to code polytopes form a mature neighboring theory [14]; no novelty is claimed for the underlying linear-algebraic viewpoint.

8 Hidden synergy

The synchronous constraints do not force different time tables to be independent.

8.1 A general interaction generator

Let γ be any one-time coupling on the full counterfactual state $R = (R(a))_{a \in \mathcal{A}}$ with the required coordinate marginals. Suppose there exists a bounded nonzero function $h(R)$ satisfying

$$\mathbb{E}_\gamma[h] = 0, \quad \mathbb{E}_\gamma[h \mid R(a)] = 0 \quad \forall a \in \mathcal{A}. \quad (8.1)$$

Proposition 8.1 (Hidden-interaction perturbation). *For sufficiently small nonzero ε , the two-time law*

$$\Gamma_\varepsilon(r, s) = \gamma(r)\gamma(s)(1 + \varepsilon h(r)h(s)) \quad (8.2)$$

is synchronous policy-safe, has both one-time marginals equal to γ , and satisfies

$$I(R_1; R_2) > 0.$$

Consequently

$$H(R_1, R_2) < 2H(\gamma). \quad (8.3)$$

Proof. Nonnegativity holds for $|\varepsilon|$ small enough. Summing over the unobserved coordinates of r and s factorizes the perturbation term into conditional means of h , which vanish by (8.1). Hence every cross-time selected coordinate pair remains a product with the correct marginals. The joint table law differs from $\gamma \otimes \gamma$ for nonzero ε , so its mutual information is positive. Since the one-time marginals remain γ , entropy drops by exactly that mutual information. $\square$

This proposition explains why hidden synergy is generic whenever the one-time coupling has a nontrivial pure interaction direction. It does *not* by itself prove a net saving relative to $2C_1$: the chosen γ may have entropy above the one-step minimum.

8.2 An improved explicit two-step witness

For the crash-test pair, let

$$R_t = (F_t(P), F_t(Q)) \in \{0, 1, 2\}^2.$$

Appendix A gives a sparse 9×9 law $\tilde{G}$ with rows indexed by R_1 and columns by R_2 .

Proposition 8.2 (Improved 23-atom witness). *At the value of q in (6.2), $\tilde{G}$ is a probability law with exactly 23 positive atoms, and for every $a, b \in \{P, Q\}$,*

$$(F_1(a), F_2(b)) \sim P_a \otimes P_b. \quad (8.4)$$

Thus $\tilde{G}$ is synchronous policy-safe. Numerically,

$$H(\tilde{G}) = 3.878295576158564 \dots \text{ bits}. \quad (8.5)$$

Proof. The entries in Appendix A sum symbolically to one. All four 3×3 transversal projections are polynomial identities in q . The numerical value in (6.2) lies in $(0.41003, 0.41004)$, on which every displayed nonzero entry is positive; the smallest is approximately 1.3278861×10^{-3} . Entropy evaluation gives (8.5). $\square$

The one-time table entropies and temporal mutual information are

$$H(R_1) = 2.860191813676764 \dots, \quad H(R_2) = 2.841039077264926 \dots, \quad (8.6)$$

$$I(R_1; R_2) = 1.822935314783125 \dots \text{ bits}. \quad (8.7)$$

More strikingly,

$$I(F_1(P); R_2) = 0.617663017691050 \dots \text{ bits}, \quad (8.8)$$

while

$$I(F_1(P); F_2(P)) = I(F_1(P); F_2(Q)) = 0. \quad (8.9)$$

An analogous statement holds after choosing Q at the first time. Thus the realized past can carry information about the *joint* future counterfactual table while carrying no information about any individually selectable future coordinate.

Corollary 8.3 (Strict block subadditivity). *For the crash test,*

$$C_2^{\text{syn}} \leq H(\tilde{G}) < 2C_1, \quad (8.10)$$

with certified feasible saving

$$2C_1 - H(\tilde{G}) = 0.229541595055955 \dots \text{ bits.} \quad (8.11)$$

Hence

$$C_\infty^{\text{syn}} \leq 1.939147788079282 \dots \text{ bits.} \quad (8.12)$$

No global optimality claim is made for $\tilde{G}$. The finite problem remains a concave entropy minimization over a marginal polytope.

8.3 A variational identity for net hidden synergy

For any synchronous policy-safe law let $R_t = (F_t(a))_{a \in \mathcal{A}}$ be its complete one-time counterfactual table. Define temporal total correlation

$$\text{TC}(R_{1:n}) := \sum_{t=1}^n H(R_t) - H(R_{1:n}). \quad (8.13)$$

Since $H(R_t) \geq C_1$ for every t ,

$$H(R_{1:n}) = nC_1 + \sum_{t=1}^n (H(R_t) - C_1) - \text{TC}(R_{1:n}). \quad (8.14)$$

Taking the minimum gives the exact identity

$$\boxed{nC_1 - C_n^{\text{syn}} = \max_{\Gamma \text{ policy-safe}} \left[\text{TC}_\Gamma(R_{1:n}) - \sum_t (H_\Gamma(R_t) - C_1) \right]}. \quad (8.15)$$

Thus net hidden synergy is not merely temporal dependence. It is temporal dependence minus the extra one-time entropy required to make that dependence possible.

9 The old overlap kernel reappears

An earlier author manuscript, *Scale Reduction for Compactly Supported Polynomial Kernels*, considered

$$\rho_k(x) = \frac{k+1}{2}(1 - |x|)_+^k, \quad \rho_{k,a}(x) = \frac{1}{a}\rho_k(x/a), \quad (9.1)$$

and the overlap

$$F_k(a, b) = \int_{\mathbb{R}} \rho_{k,a}(x) \rho_{k,b}(x) dx. \quad (9.2)$$

For $0 < a \leq b$ it derived

$$F_k(a, b) = \frac{1}{b} H_k(a/b), \quad (9.3)$$

with an explicit polynomial/hypergeometric H_k and alternating derivative signs. We use only the elementary $k = 0$ case here [24].

For $k = 0$, $\rho_{0,a}$ is the uniform density on $[-a, a]$, hence

$$F_0(a, b) = \frac{1}{2 \max(a, b)}. \quad (9.4)$$

Under the reciprocal probability-scale substitution

$$a = \frac{1}{p}, \quad b = \frac{1}{q},$$

we obtain the exact identity

$$\boxed{2F_0(1/p, 1/q) = \min(p, q)}. \quad (9.5)$$

If

$$u = -\log_2 p, \quad v = -\log_2 q,$$

then

$$2F_0(2^u, 2^v) = 2^{-\max(u, v)},$$

and therefore

$$\boxed{-\log_2(2F_0(2^u, 2^v)) = \max(u, v)}. \quad (9.6)$$

This is exactly the local scalar cost that appears in the counterfactual information-spectrum converse below.

Remark 9.1 (What this does and does not mean). Equation (9.5) is an identity, not an analogy. It does *not* mean that the earlier kernel note contained the policy-safe theorem: that note had no transversal constraints, block entropy, hidden synergy, or optimal transport over surprisal laws. It supplied an overlap representation of the scalar minimum. The present argument explains why that minimum becomes an information bill.

For general k , writing $m = \min(p, q)$ and $M = \max(p, q)$ gives

$$F_k(1/p, 1/q) = m H_k(m/M) = 2^{-\max(u, v)} H_k(2^{-|u-v|}). \quad (9.7)$$

The higher- k family may be analytically suggestive, but no stronger counterfactual converse is claimed from it.

10 Information-spectrum geometry

10.1 Two actions

For two actions write

$$X_t := F_t(P), \quad Z_t := F_t(Q),$$

and define surprisals

$$\iota_P(x) := -\log_2 P(x), \quad \iota_Q(z) := -\log_2 Q(z).$$

Let μ_P and μ_Q be the laws of $\iota_P(X)$ and $\iota_Q(Z)$ under their respective marginals.

Theorem 10.1 (Two-action synchronous information-spectrum bound). *Every synchronous policy-safe n -block satisfies*

$$H(F) \geq \frac{n}{2} \left(H(P) + H(Q) + W_1(\mu_P, \mu_Q) \right). \quad (10.1)$$

Consequently

$$C_\infty^{\text{syn}}(P, Q) \geq \frac{1}{2} \left(H(P) + H(Q) + W_1(\mu_P, \mu_Q) \right). \quad (10.2)$$

Proof. Fix a positive-probability full-table atom

$$f = (x_1, z_1, \dots, x_n, z_n)$$

with probability $\Gamma(f)$. For every subset $S \subseteq [n]$, this atom lies inside the transversal event selecting P on S and Q on S^c . Hence

$$\Gamma(f) \leq \prod_{t \in S} P(x_t) \prod_{t \notin S} Q(z_t).$$

Minimizing over S separates coordinatewise:

$$\Gamma(f) \leq \prod_t \min\{P(x_t), Q(z_t)\}. \quad (10.3)$$

Using (9.5), this can equivalently be written as a product of $k = 0$ overlap kernels. Taking $-\log_2$ gives

$$-\log_2 \Gamma(f) \geq \sum_t \max\{\iota_P(x_t), \iota_Q(z_t)\}. \quad (10.4)$$

Average over F . At each time (X_t, Z_t) induces some coupling of P, Q , hence its surprisal pair induces a coupling of μ_P, μ_Q . Since

$$\max(u, v) = \frac{1}{2}(u + v + |u - v|),$$

minimizing the expected absolute difference gives W_1 . □

If $H(P) = H(Q) = h$, then

$$C_\infty^{\text{syn}}(P, Q) \geq h + \frac{1}{2}W_1(\mu_P, \mu_Q). \quad (10.5)$$

Moreover $W_1(\mu_P, \mu_Q) = 0$ exactly when the nonzero probability spectra agree up to permutation: the surprisal mass at $-\log_2 r$ equals the multiplicity of atom size r times r .

10.2 Arbitrary finite action families

The two-action W_1 formula is one slice of a simpler multi-action geometry.

For each $a \in \mathcal{A}$, let μ_a be the law of

$$U_a := -\log_2 P_a(Y_a), \quad Y_a \sim P_a.$$

Define

$$T(P_a : a \in \mathcal{A}) := \inf_{\eta \in \Pi(\mu_a : a \in \mathcal{A})} \mathbb{E}_\eta \max_{a \in \mathcal{A}} U_a. \quad (10.6)$$

Lemma 10.2 (Quantile envelope). *For nonnegative finite-support laws μ_a ,*

$$\boxed{T(P_a : a \in \mathcal{A}) = \int_0^1 \max_{a \in \mathcal{A}} F_{\mu_a}^{-1}(u) du.} \quad (10.7)$$

Proof. For any coupling,

$$\Pr(\max_a U_a \leq t) = \Pr(U_a \leq t \ \forall a) \leq \min_a F_{\mu_a}(t).$$

The comonotone coupling

$$U_a = F_{\mu_a}^{-1}(V), \quad V \sim \text{Unif}[0, 1],$$

attains equality simultaneously for every t . Therefore it stochastically minimizes the maximum, and integration of the quantile of the maximum gives (10.7). □

Theorem 10.3 (Multi-action synchronous information-spectrum bound). *For any finite action family and any synchronous policy-safe block,*

$$C_n^{\text{syn}}(P_a : a \in \mathcal{A}) \geq n \int_0^1 \max_{a \in \mathcal{A}} F_{\mu_a}^{-1}(u) du. \quad (10.8)$$

Proof. A full-table atom must fit inside every action-string transversal. Minimizing the corresponding product probability over action strings separates over time and yields

$$\Gamma(f) \leq \prod_t \min_a P_a(f_{t,a}).$$

Thus

$$-\log_2 \Gamma(f) \geq \sum_t \max_a [-\log_2 P_a(f_{t,a})].$$

At each time the surprisal vector has a coupling of the prescribed μ_a marginals. Lemma 10.2 supplies the least possible expected maximum. $\square$

If all actions have equal entropy h , then the right side of (10.8) is at least nh . Equality holds only if all surprisal laws coincide: equal means plus equality of the expected maximum force all comonotone quantiles to agree almost everywhere. Thus a mismatch of any two surprisal spectra creates a positive synchronous asymptotic surcharge.

11 The crash test after the improved bounds

For

$$P = (1/2, 1/4, 1/4),$$

the surprisal law is

$$\mu_P = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_2. \quad (11.1)$$

For Q ,

$$\mu_Q = 2q \delta_{-\log_2 q} + (1 - 2q) \delta_{-\log_2(1-2q)}. \quad (11.2)$$

Numerically,

$$W_1(\mu_P, \mu_Q) = 0.4569309374910615 \dots,$$

so Theorem 10.1 gives

$$C_\infty^{\text{syn}} \geq 1.7284654687455308 \dots \quad (11.3)$$

Combining this with Corollary 8.3 gives the updated certified interval

$$1.728465468745531 \leq C_\infty^{\text{syn}} \leq 1.939147788079282. \quad (11.4)$$

At finite horizon two, Proposition 7.3 and the explicit witness give

$$3.553918585607260 \leq C_2^{\text{syn}} \leq 3.878295576158564. \quad (11.5)$$

The upper endpoint is a feasible vertex value, not a certified global optimum.

12 Counterfactual breadth is a geometry, not a count

Full synchronous policy-safety requires every action string. To interpolate between one realized history and a complete rule, let

$$\mathcal{G} \subseteq \{P, Q\}^n$$

be only the family of action strings whose transversal laws must be correct. Let $C_n(\mathcal{G})$ be the minimum full-table entropy under those constraints.

For a full atom define

$$M_t := \max\{\iota_P(X_t), \iota_Q(Z_t)\}, \quad \Delta_t := |\iota_P(X_t) - \iota_Q(Z_t)|,$$

and let $s \in \{P, Q\}^n$ choose at each time the action attaining M_t . For $g \in \mathcal{G}$ define weighted Hamming distance

$$d_\Delta(s, g) := \sum_{t: s_t \neq g_t} \Delta_t. \quad (12.1)$$

Then the exact pointwise converse is

$$\boxed{-\log_2 \Gamma(f) \geq \sum_t M_t - d_\Delta(s, \mathcal{G})}, \quad (12.2)$$

where

$$d_\Delta(s, \mathcal{G}) := \min_{g \in \mathcal{G}} d_\Delta(s, g).$$

Thus breadth is governed by how well the required action family covers the locally most expensive action strings, not merely by $|\mathcal{G}|$.

Assume both action marginals are required at every coordinate and both distributions have positive support. Let

$$D := \max_{x, z} |\iota_P(x) - \iota_Q(z)|,$$

and let

$$r(\mathcal{G}) := \max_s \min_{g \in \mathcal{G}} d_H(s, g)$$

be ordinary Hamming covering radius.

Corollary 12.1 (Covering-radius converse). *With*

$$T(P, Q) = \frac{1}{2}(H(P) + H(Q) + W_1(\mu_P, \mu_Q)),$$

we have

$$\boxed{C_n(\mathcal{G}) \geq nT(P, Q) - D r(\mathcal{G})}. \quad (12.3)$$

In particular, if $r(\mathcal{G}_n) = o(n)$, the full synchronous information-spectrum rate survives asymptotically up to $o(1)$ per step.

This shows why cardinality alone is misleading. The even-parity subset of $\{P, Q\}^n$ contains only half the action strings but has covering radius one, so the full tax is lost by at most an $O(1)$ term. Conversely, genuinely narrow families can have extensive covering radius.

A particularly transparent family is a k -dimensional subcube: actions are free on k coordinates and fixed on the other $n - k$. If $H(P) = H(Q) = h$, direct factorization of the atom bound gives

$$\boxed{C_n(\mathcal{G}) \geq nh + k(T(P, Q) - h)}. \quad (12.4)$$

Thus each independently branching intervention site contributes the same local surcharge in this model. History corresponds to $k = 0$; a rule with extensive independent branching corresponds to $k = \Theta(n)$.

13 From block source entropy to innovation rate

The synchronous quantity C_n^{syn} is an entropy of a precommitted source that fixes the complete stable-event table. It must not be silently identified with every possible notion of dynamic fresh randomness.

Suppose that for each horizon

$$F^{(n)} = \phi_n(S_0, U_{1:n}), \quad (13.1)$$

where S_0 is an initial reservoir independent of the external innovation source. No Markov assumption is needed.

Proposition 13.1 (Finite-preload lower bridge). *Every implementation of the form (13.1) satisfies*

$$H(U_{1:n}) \geq C_n^{\text{syn}} - H(S_0). \quad (13.2)$$

If $H(S_0) = o(n)$, its asymptotic innovation entropy rate is at least C_∞^{syn} .

Proof. Since $F^{(n)}$ is deterministic given $(S_0, U_{1:n})$,

$$C_n^{\text{syn}} \leq H(F^{(n)}) \leq H(S_0, U_{1:n}) \leq H(S_0) + H(U_{1:n}).$$

□

For an entropy-priced precommitted source the converse is also trivial in the other direction: take $U = F^{(n)}$ with an optimal block law and concatenate blocks. Hence, under this explicit accounting standard,

$$\boxed{R_{\text{precommitted}} = C_\infty^{\text{syn}}}. \quad (13.3)$$

If innovation is priced instead as expected fair-coin consumption, exact random-number generation introduces coding overhead; modern recycling results show that Shannon entropy can be approached with sublinear block overhead in closely related sampling settings [8, 9]. Multi-letter entropy rates are also standard in exact common-information and exact channel-synthesis problems [5, 6].

14 History-indexed causal trees: the real remaining boundary

The synchronous table $F_t(a)$ imposes a stable identity across histories. A more general causal contingent plan may use

$$F_t(h_{t-1}, a), \quad (14.1)$$

where h_{t-1} records previous actions and outputs. This distinction is close in spirit to contingent plans in extensive-form games and to filtration-compatible coupling in causal transport [11, 12, 15].

Definition 14.1 (Causal-tree entropy). Let $C_n^{\text{causal}}(W)$ be the minimum Shannon entropy of a finite random history-indexed response tree whose induced realized process is exactly the memoryless channel under every adaptive nonanticipative policy.

At horizon one, no history exists, so

$$C_1^{\text{causal}} = C_1. \quad (14.3)$$

Every synchronous table induces a causal tree by setting $F_t(h, a) = F_t(a)$, hence

$$C_n^{\text{causal}} \leq C_n^{\text{syn}}. \quad (14.4)$$

Proposition 14.2 (Causal prefix recursion). *For every $n \geq 1$,*

$$\boxed{C_n^{\text{causal}} \geq C_1 + (n-1)h_{\max}.} \quad (14.5)$$

Moreover C_n^{causal} is subadditive, so C_∞^{causal} exists and

$$\boxed{h_{\max} \leq C_\infty^{\text{causal}} \leq C_\infty^{\text{syn}}.} \quad (14.6)$$

Proof. Let T be any exact horizon- n causal tree. Choose an entropy-maximizing root action a_* . The pair consisting of its root output Y and the realized child subtree T_Y is a deterministic function of T , so

$$H(T) \geq H(Y, T_Y) = H(P_{a_*}) + H(T_Y | Y).$$

Conditioned on each positive-probability root output, the child subtree must again exactly realize the same memoryless channel for every future policy; otherwise a policy that chooses a_* first and then the offending continuation would distinguish the construction. Therefore

$$H(T_Y | Y = y) \geq C_{n-1}^{\text{causal}}$$

for every such y , giving

$$C_n^{\text{causal}} \geq h_{\max} + C_{n-1}^{\text{causal}}.$$

Iteration yields (14.5). Independent block concatenation gives subadditivity and hence the asymptotic limit. $\square$

This is where the strongest philosophical claim stops. We have proved a positive asymptotic surcharge for C3-S, not for unrestricted C3-C.

Open Problem 14.3 (Causal-tree tax). For the equal-entropy crash-test pair, determine whether

$$C_\infty^{\text{causal}} > 1.5.$$

More generally, characterize when stable history-indexing can lower the asymptotic cost below C_∞^{syn} and whether any nonzero counterfactual surcharge survives after that freedom is allowed.

A negative answer, $C_\infty^{\text{causal}} = h_{\max}$, would mean that the positive theorem above prices *stable cross-history event identity*, not counterfactuality by itself. A positive answer would establish a genuinely broader causal counterfactual tax.

15 Precision, memory, and stable event identity

A scalar entropy rate is still not the whole bill. An exact real-valued state can hide an infinite discrete tape in its binary expansion, so precision or description complexity must be priced if exact real states are admitted. Persistent memory can substitute for fresh randomness, as already seen in near-Shannon samplers [10]. Stable event identity can itself be weakened by moving from C3-S to C3-C. A useful resource geometry therefore remains at least

$$(M, R, \Pi, B, I), \quad (15.1)$$

where M is persistent memory, R fresh innovation, Π precision or description complexity, B counterfactual breadth, and I records the strength of cross-history identity imposed on latent events.

This fifth coordinate is the main conceptual correction of Version 3.

16 What is proved, what is not, and what remains open

Proved under the synchronous stable-event contract

- Transversal correctness is equivalent to exactness for every adaptive nonanticipative policy within the synchronous table model.
- C_n^{syn} is subadditive, has a well-defined asymptotic rate, and satisfies the prefix lower bound (7.5).
- The transversal marginal polytope has rank d_{obs}^n with $d_{\text{obs}} = 1 + \sum_a (d_a - 1)$.
- Hidden interaction can be generated by zero-conditional-mean perturbations, and the crash test exhibits strict block subadditivity with an explicit 23-atom witness.
- Net hidden synergy has the exact variational identity (8.15).
- The local probability minimum in the converse is exactly the reciprocal-scale $k = 0$ overlap kernel of the earlier compact-support note.
- The two-action W_1 lower bound extends to arbitrary finite action families via the quantile-envelope theorem (10.8).
- Counterfactual breadth admits a covering-geometric converse; sublinear covering radius preserves the full synchronous tax asymptotically.
- For entropy-priced precommitted sources with sublinear preload, the innovation rate equals C_∞^{syn} .

Not proved

- The 23-atom witness is not claimed to minimize C_2^{syn} .
- The exact value of C_∞^{syn} for the crash test is unknown.
- No theorem here proves $C_\infty^{\text{causal}} > h_{\text{max}}$ for unrestricted history-indexed causal trees.
- No entropy inequality here identifies the physical substrate, disproves simulation, or decides whether observed randomness is ontic or epistemic.
- No historical novelty is claimed for the terminology, kernels, interaction decompositions, or derived inequalities.

Open Problem 16.1 (Exact synchronous asymptotic rate). For the crash-test pair, determine

$$C_\infty^{\text{syn}} \in [1.728465468745531, 1.939147788079282].$$

Is the quantile/transport converse sharp, or do simultaneous transversal compatibility constraints generate a stronger asymptotic invariant?

Open Problem 16.2 (Exact finite two-step entropy). Determine C_2^{syn} exactly. The current certified bracket is

$$3.553918585607260 \leq C_2^{\text{syn}} \leq 3.878295576158564.$$

The upper endpoint is the explicit vertex in Appendix A.

Open Problem 16.3 (Sharp breadth–rate geometry). For general action-string families $\mathcal{G}_n$, characterize the asymptotic source rate in terms of intrinsic covering or coding geometry. Cardinality alone is insufficient.

Open Problem 14.3 is the most important conceptual boundary: does the tax survive after stable cross-history event identity is removed?

17 Substrate skepticism after the mathematics

The mathematics now supports a narrower and cleaner philosophical statement than Version 2.

A history is what happened. A rule constrains what would happen under alternatives. Under the synchronous contract, those alternatives are not merely separately valid: the same time/action latent event is required to have a stable identity across mutually exclusive histories. That compatibility has a positive asymptotic source cost whenever self-information spectra differ.

Hidden correlations can reduce the price. They cannot erase it under C3-S. But a general history-indexed causal tree is a weaker contract, and its asymptotic tax remains unresolved.

Thus the strongest defensible statement is

We do not know what reality is made of. We can, however, price explicit notions of counterfactual consistency. If mutually exclusive histories must share a stable synchronous latent identity, equal observable Shannon entropy is not enough: a mismatch of self-information spectra leaves a positive asymptotic source bill. Whether counterfactual coherence without that stable identity still carries a positive tax remains open.

The old question therefore survives, but more precisely:

Who pays, and which notion of “the same unrealized event” are we charging for?

18 Conclusion

The attack changed the target repeatedly. The one-shot MEC surcharge was too large to be interpreted naively as a per-step tax. Hidden synergy showed why: mutually exclusive futures can share latent information. The information-spectrum converse then showed that, under a synchronous stable-event contract, such netting cannot remove the entire mismatch. The improved crash-test interval is

$$1.5 < 1.7284654687 \leq C_{\infty}^{\text{syn}} \leq 1.9391477881 < 2.0539185856 = C_1.$$

The two-action Wasserstein formula is only the first member of a multi-action quantile-envelope geometry. The same local minimum is also exactly the $k = 0$ reciprocal-scale overlap kernel from an earlier compact-support calculation:

$$\min(p, q) = 2F_0(1/p, 1/q).$$

What looked like a coincidence is a shared lattice operation: reciprocal scale turns maximum support width into minimum probability, and logarithms turn that minimum into maximum surprisal.

The largest remaining conceptual question is no longer hidden synergy. It is the gap between synchronous stable-event tables and unrestricted history-indexed causal trees. A positive synchronous tax is proved. A positive causal-tree tax is not.

The one-shot bill was too large. The Shannon bill was too small. Hidden correlations pay part of the difference. Stable identity appears to pay another part. The final accountant remains unidentified.

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imply endorsement of the framework, terminology, mathematics, or conclusions by Dr. Arazim or by the project.

The author is not writing from an academic position in information theory, probability, causal inference, or mathematical physics. The investigation is amateur in the literal sense: it is pursued because the questions are interesting. The standing assumption remains that any elegant theorem encountered along the way probably has a prior owner whom we have not yet found.

A The improved 23-atom witness

Order one-time states by

$$\mathcal{R} = (00, 01, 02, 10, 11, 12, 20, 21, 22),$$

where the first digit is the P coordinate and the second the Q coordinate. Rows index R_1 and columns index R_2 . The witness is

$$\tilde{G} = \begin{pmatrix} \frac{(2q-1)(12q-5)}{8} & 0 & -\frac{q(2q-1)}{8} & 0 & 0 & 0 & 0 & \frac{q}{4} & 0 \\ q^2 & 0 & -\frac{(2q-1)(8q-3)}{8} & 0 & \frac{q}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2q-1}{8} & 0 & 0 & -\frac{(2q-1)(8q-3)}{4} & 0 & \frac{(2q-1)(16q-7)}{8} \\ \frac{q}{4} & 0 & 0 & 0 & \frac{4q-1}{16} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2q-1}{8} & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2q-1}{8} & 0 & 0 & 0 \\ -\frac{32q^2-40q+11}{16} & \frac{(4q-1)^2}{16} & 0 & \frac{1}{16} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{16q^2-4q-1}{16} & -\frac{12q-5}{16} & 0 & 0 & 0 & 0 & 0 & \frac{4q-1}{16} \\ \frac{(2q-1)^2}{2} & -q(2q-1) & 0 & 0 & 0 & 0 & -\frac{2q-1}{8} & 0 & 0 \end{pmatrix}. \quad (\text{A.1})$$

The four transversal projections are polynomial identities in q and do not use the entropy equation (6.2); that equation is required only to select the equal-entropy crash-test value.

B Reproducibility checks

The companion script `wpb_v3_verify.py`:

1. solves (6.2) numerically and verifies $H(P) = H(Q) = 1.5$;
2. evaluates the one-shot MEC witness and C_1 ;
3. verifies symbolically that the 23 entries of $\tilde{G}$ sum to one;
4. verifies symbolically all four 3×3 transversal projections;
5. checks positivity and counts the 23 nonzero atoms;
6. evaluates $H(\tilde{G})$, one-time table entropies, and hidden mutual informations;
7. evaluates $W_1(\mu_P, \mu_Q)$ and the synchronous asymptotic lower bound;
8. checks the reciprocal-scale $k = 0$ overlap identity in the two possible orderings of p and q .

All numerical constants quoted for the crash test are regenerated by that script.

C Literature map rather than novelty claim

Topic	Status used here	Representative neighbor
Minimum-entropy coupling	Known	Kovačević–Stanojević–Šenk [2]

Information-spectrum MEC converses	Known	Shkel–Yadav [3]
Approximate multi-marginal MEC	Known	Li [4]
Exact common-information rate	Known	Kumar–Li–El Gamal [5]
Exact channel synthesis	Known	Yu–Tan [6]
Joint entropy under structured independence	Known	Gavinsky–Pudlák [13]
Marginal polytopes and codes	Known	Kahle–Wenzel–Ay [14]
Chosen-path randomness recycling	Known	Draper–Saad; Draper–Harris–Saad [8, 9]
Finite-memory mixed/behavioral strategies	Known	Main–Randour [12]
Causal/bicausal transport	Known	Cont–Lim [15]
Synergistic disclosure / sample privacy	Known	Rassouli–Rosas–Gündüz [16]
Stable event-keyed common randomness	Known neighboring issue	Buffalo–Pearson–Klein [23]
Compact-support overlap family F_k	Earlier author note	Mikulík [24]; only the $k = 0$ identity is used here
Synchronous block entropy C_n^{syn}	Not located in this formulation	No direct source located in the present search
Multi-action quantile-envelope converse under all synchronous transversals	Not located in this formulation	Compare MEC information-spectrum converses [3]
Causal-tree entropy C_n^{causal} and its gap from C_n^{syn}	Open here	Compare contingent plans and causal transport [12, 15]

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